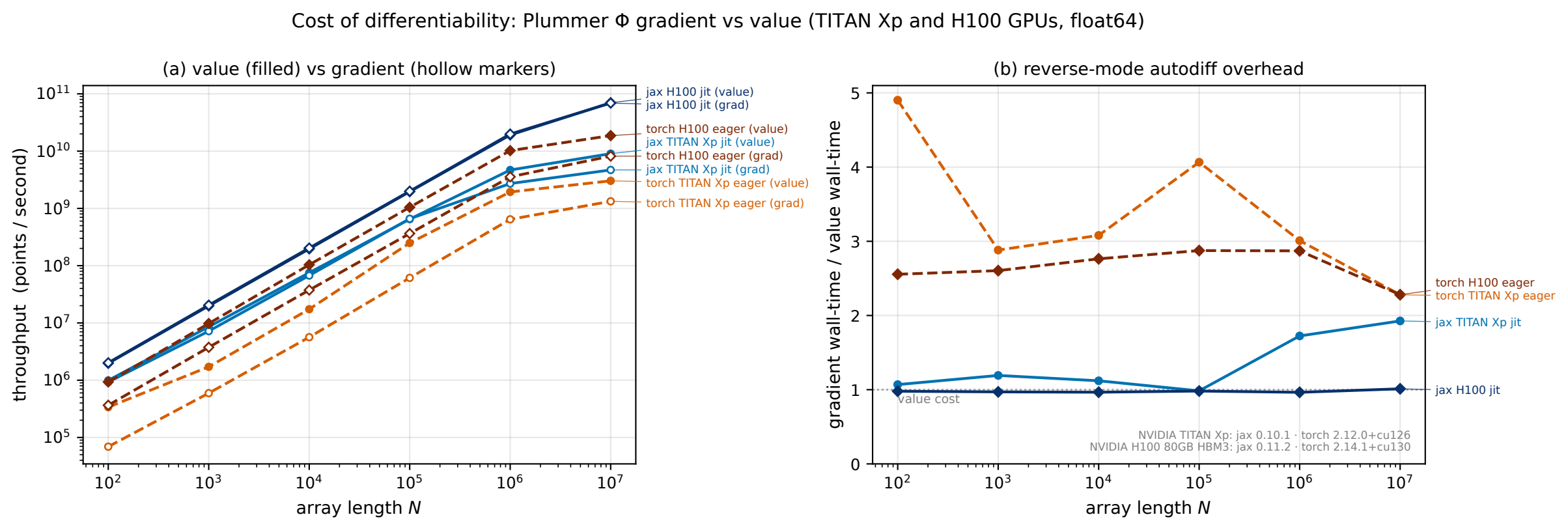
Cost of differentiability: Plummer Φ gradient vs value (TITAN Xp and H100 GPUs, float64)
(a) value (filled) vs gradient (hollow markers)
throughput (points / second)
array length N
jax H100 jit (value)
jax H100 jit (grad)
torch H100 eager (value)
jax TITAN Xp jit (value)
torch H100 eager (grad)
jax TITAN Xp jit (grad)
torch TITAN Xp eager (value)
torch TITAN Xp eager (grad)
(b) reverse-mode autodiff overhead
gradient wall-time / value wall-time
array length N
value cost
torch H100 eager
torch TITAN Xp eager
jax TITAN Xp jit
jax H100 jit
NVIDIA TITAN Xp: jax 0.10.1 · torch 2.12.0+cu126
NVIDIA H100 80GB HBM3: jax 0.11.2 · torch 2.14.1+cu130